Green University of Bangladesh
Department of Computer Science and Engineering(CSE)

Faculty of Sciences and Engineering

Semester: (Fall, Year:2024), B.Sc. in CSE (Day)

LAB REPORT NO :7

Course Title:Microprocessor and Microcontroller Lab

Course Code:Cse 304 Section:222 D3

Lab Experiment Name: Implement Macro in Assembly Language Programming

Student Details

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Lab Date :28/11/2024

Submission Date : 29/11/2024

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Lab Report Status

Marks: …… Signature:.....

Comments:..... Date:.....

1. TITLE OF THE LAB EXPERIMENT

Implementing Macros for Code Reusability in Assembly Language.

2. OBJECTIVES/AIM

- Demonstrate array handling and macro usage in Assembly Language.
- Calculate and separate the sums of even and odd digits within an array.
- Explore techniques for code reusability and modularity in Assembly.
- Apply Assembly concepts to solve a practical programming task.

PROCEDURE

Initialize data: Set the data segment to point to the data section.Load the array address and size into registers.Initialize variables evenSum and oddSum to zero.

Iterate through the array: Use a loop to process each element of the array.Load the current element into the AL register.

Process each digit: Call the processDigit macro to determine if the digit is even or odd and add it to the appropriate sum.

Macros: processDigit:
Checks if the digit in AL is even or odd.
Adds the digit to evenSum if even, or oddSum if odd.
printDigit:
Converts a digit in DL to its ASCII character.
Prints the character to the console.

Display results: Print a message indicating the results.

4. IMPLEMENTATION

```
.model small
.data
   array DB 1, 2, 3, 4, 5, 6, 7, 8, 9
```

```
  arraySize DW 9 ; Size of the array
    evenSum DB ?
    oddSum DB ?
.code
    main PROC
        mov ax, @data
        mov ds, ax
        lea si, array ; SI points to the beginning of the array
        mov cx, arraySize ; CX is the loop counter
        mov evenSum, 0 ; Initialize even sum to zero
        mov oddSum, 0 ; Initialize odd sum to zero
    arrayLoop:
        mov al, [si]
        inc si
          call processDigit
        loop arrayLoop
            mov ah, 09h
        lea dx, resultMsg
        int 21h
        mov dl, evenSum
        call printDigit

        mov ah, 09h
        lea dx, oddMsg
        int 21h

        mov dl, oddSum
        call printDigit

        mov ah, 4Ch
        int 21h

    main ENDP

    processDigit MACRO

                test al, 1
        jz evenDigit
        add oddSum, al
        jmp endProcess
    evenDigit:
        add evenSum, al ; Add the even digit to evenSum
    endProcess:
    ENDM
    printDigit MACRO
       add dl, '0'
```

```
    mov ah, 02h
    int 21h
ENDM
resultMsg DB 'Results:', 0
oddMsg DB 'Sum of odd digits: ', 0
END main
```

TEST RESULT / OUTPUT

```
Enter the size of the array: 4
Enter 4 elements:
Enter element 1: 2
Enter element 2: 5
Enter element 3: 6
Enter element 4: 7
Sum of even DIGIT: 8
Sum of odd DIGIT: 12
```

ANALYSIS AND DISCUSSION

Calculates and separately displays the sum of even and odd digits within a given array.Employs macros for modularity, leverages efficient bitwise operations for odd/even checks, and utilizes DOS interrupts for output.Initializes variables, iterates through the array using a loop, processes each digit using a macro, prints results, and exits.Could incorporate a procedure for printing numbers in decimal format for more versatile output.